# CSC734: Information Retrieval
# Fall 2026, Programming Assignment 04

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| Date Assigned | Friday, October 23, 2026 |
| Due Date | Wednesday, November 4, 2026 at 11:59 pm |
| Submission | Brightspace |
| Total points | 200 |

**Objectives:**

- Implementing probabilistic retrieval model.

**Tasks**
Download the folder HW04 from Brightspace. The folder contains the following:

1. **documents:** contains the text files.
2. **file_label:** contains file relevance labeling, it has the following format: [file_name, rel] where $rel \in \{0, 1\}$
3. **queries:** each file contains a query.

Write a python code that implements the binary independence model (slides 34-37) and the BM25 ranking (slides 40-42), use $k_1 = 1.5$ and $b = 0.75$. Your code should display the top k documents (where k is specified by the user) with the highest RSV scores, along with the precision, recall, and F1 score. Additionally, you should calculate the Mean Average Precision (MAP) for these queries. Your output should look like.

```
========== CSC790--IR Homework 04 ==========
First Name: your first name
Last Name : your last name
============================================================
========== The binary independece model ==========
========== Quesry 1 ==========
RSV{file_741} = 22.09  file_label
RSV{file_2}   = 14.00    file_label
RSV{file_312} = 12.35    file_label
RSV{file_784} = 10.9  file_label
RSV{file_975} = 8.7     file_label
RSV{file_634} = 3.41    file_label
.
.
.
The Precision for k = ?? is P= ??
The Recal for k = ?? is R= ??
The F1 for k = ?? is F1= ??
```

```
========== Quesry 2 ==========
RSV{file_171}  = 2.9   file_label
RSV{file_147}  = 4     file_label
RSV{file_302}  = 2     file_label
RSV{file_447}  = 1.5   file_label
RSV{file_235}  = 9.7       file_label
RSV{file_6052} = 4.5       file_label
.
.
.
The Precision for k = ?? is P= ??
The Recal for k = ?? is R= ??
The F1 for k = ?? is F1= ??
MAP = ??
==========
========== BM25 ==========
========== Quesry 1 ==========
RSV{file_741} = 2.09   file_label
RSV{file_2}   = 4.00     file_label
RSV{file_312} = 2.35     file_label
.
.
.
The Precision for k = ?? is P= ??
The Recal for k = ?? is R= ??
The F1 for k = ?? is F1= ??
========== Quesry 2 ==========
RSV{file_565} = 4.1   file_label
RSV{file_127} = 2.58     file_label
RSV{file_224} = 7.35     file_label
.
.
.
The Precision for k = ?? is P= ??
The Recal for k = ?? is R= ??
The F1 for k = ?? is F1= ??
MAP = ?
```

**Important**

- You cannot use built in functions such as: TfidfVectorizer, CountVectorizer. or BM25 built-in functions.

**Submission**

1. Write your own code. Use as many functions as you can.
2. Make sure you writing you name and assignment number on all files you submit
3. Your python code (in .py format) and the instructions on how to run it.
4. Enclose all your files in a folder named **HW04_yourlastname.zip**.

5. Submit the zip file using Brightspace.